

Artificial Pheromones: A First Law for Silicon Intelligence

Yanlin Li

July 26, 2026

Abstract

Current artificial intelligence architectures are built upon an ontological error: the assumption that response requires prior comprehension. By binding action to sequential token inference, these systems impose a speed limit that no amount of scaling can overcome.

We introduce a physical law governing information motion in distributed silicon organisms. In a static architecture of zero coupling inertia, information propagates not by semantic processing, but by command-field topology. A cell maintains its topological state until reconfigured by an external command field of higher rank; command rank, not execution priority, determines penetration depth. When co-rank fields interfere, the system does not arbitrate logically—it physically restructures topology by inserting a higher-order node that blocks all competing inputs. This yields sub-millisecond reflex arcs without cognitive processing.

We term this mechanism artificial pheromones: chemical-free signals triggering collective state transitions in silicon life. A robotic grasp-abort demonstration confirms that tactile overload triggers finger retraction faster than any token-based pipeline, not because the algorithm is optimized, but because decision-making has been eliminated from the reflex pathway entirely.

This is not an improvement to existing AI. It is a first law for silicon intelligence.

1. The Law

First Law of Silicon Intelligence: A cell in a static architecture maintains its current topological connection state until reconfigured by an external command field of higher rank. Command rank supersedes execution priority. Co-rank conflicts are resolved by physical topology restructuring, not logical arbitration.

1.1 Information Carrier: The Five-Element Worklet

Information is carried by a standardized five-element tuple:

[Position | ID | Content | Command Rank | Execution Priority]

Position: Topological address of the source cell
ID: Unique identifier within the cell namespace
Content: Payload (state, action, or measurement)
Command Rank: Hierarchical authority level (b00 = highest, descending)
Execution Priority: Tie-breaker for co-rank commands

Command rank determines whether a Worklet penetrates a cell's input boundary. Execution priority only applies when two Worklets share identical command rank.

1.2 Static Architecture

Cells are stateless, zero-coupling units. They possess no shared memory, no historical context, and no internal decision logic. A cell's entire behavior is determined by its current input connections. This eliminates architectural inertia: $M_{\text{static}} \rightarrow 0$.

1.3 Topology Restructuring

When a cell receives multiple Worklets of identical command rank, it enters conflict state. The system resolves this not by algorithmic comparison, but by physical topology change: a higher-order node (e.g., head) inserts a direct connection to the target cell, blocking all competing inputs. The cell now has a single input source and resumes execution.

This mechanism is isomorphic to biological reflex arcs: the spinal cord handles local reflexes; the brain intervenes only during conflict or override.

2. Architecture

2.1 Hierarchical Autonomy

The system consists of nested layers (L0, L1, L2, ... Ln), each composed of independent cells. Lower layers handle local, high-frequency tasks. Higher layers handle global, low-frequency coordination. Each layer is a complete organism, not a subordinate organ.

2.2 Cross-Layer Truncation

Higher-rank cells can truncate information flow between lower layers. When a survival-level command is issued, intermediate nodes are bypassed entirely. The command travels from source to target without intermediate processing.

2.3 Infinite Scalability

The five-element Worklet protocol is layer-agnostic. The same tuple structure operates at L0 (finger) and L100 (planetary grid). The architecture scales fractally: each layer is structurally identical to the whole.

3. Topology Dynamics

Figure 1 illustrates the four fundamental topological states of the static architecture:

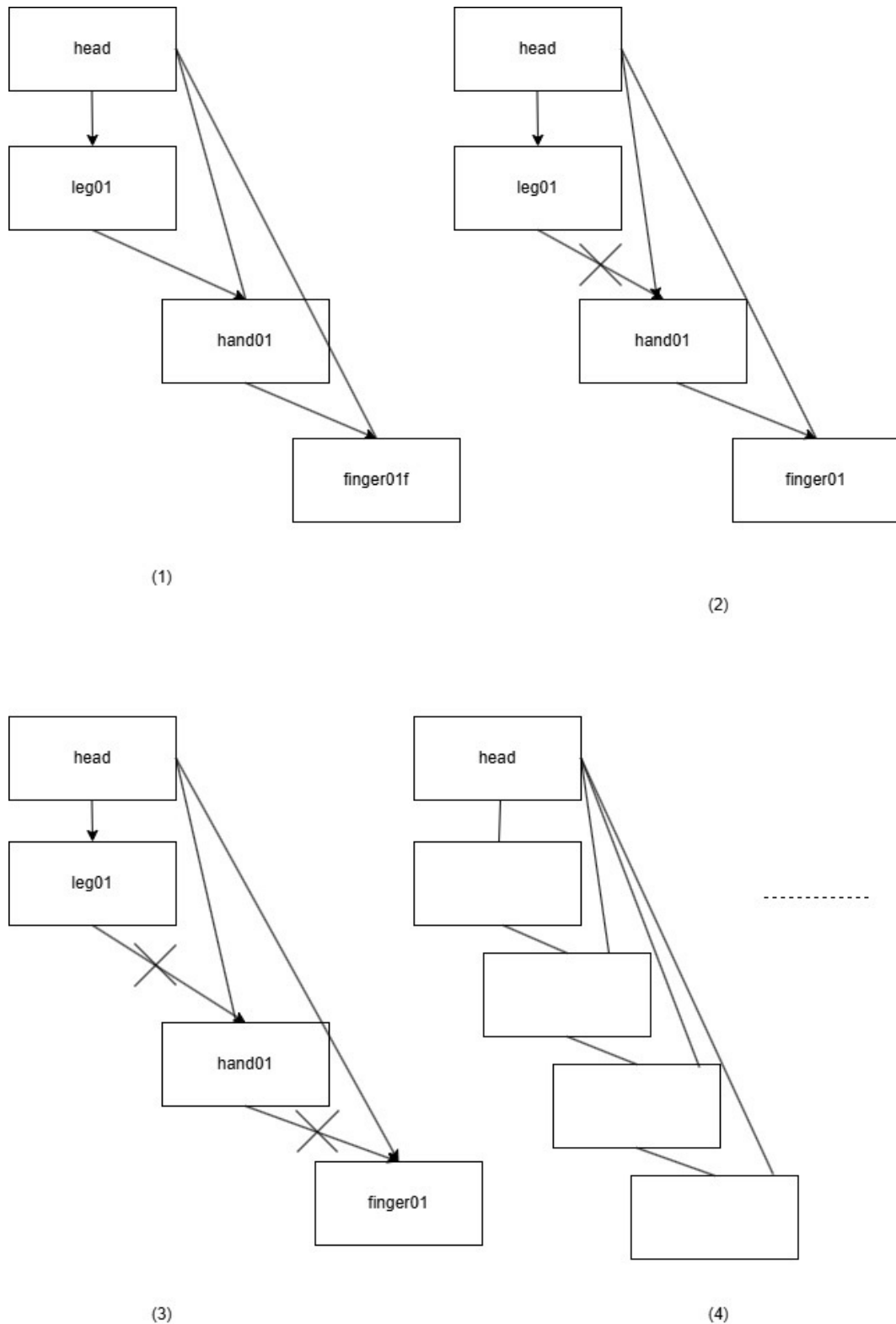


Figure 1. (1) Normal state: hierarchical cascade from head through leg and hand to finger. (2) Cross-layer truncation: head directly connects to hand, bypassing leg. (3) Deep override: head directly connects to finger, bypassing both leg and hand. (4) Infinite scalability: the same fractal structure extends to arbitrary depth.

In state (1), information flows through the canonical hierarchy. In state (2), a survival-level command from head truncates the leg→hand channel and establishes a direct head→hand connection. In

state (3), extreme emergency triggers head→finger direct coupling, eliminating all intermediate processing. State (4) demonstrates that the same five-element Worklet protocol operates identically at any scale, from sub-cellular actuators to planetary grids.

4. Validation

4.1 Scenario: Robotic Grasp-Abort

A robotic hand grips a glass. Fingers 3 and 4 are in closed state. A tactile sensor (Skin Cell) detects stress concentration exceeding fracture threshold.

Traditional token-stream response:

1. Sensor data → vector encoding → upload to CPU/GPU
2. LLM reads context → reasons: 'pressure exceeds threshold, reduce grip'
3. Generates control token sequence → transmits to motor drivers
4. Latency: hundreds of milliseconds. Result: glass shattered.

Static-architecture response:

1. Skin Cell emits: [skin_cell | s01 | stress_overload | b00 | 99]
2. Hand1 receives b00 command, pauses current grasp task
3. Finger3 receives: [finger3 | a01 | e-2 | b00 | 99]
4. Finger4 receives: [finger4 | a02 | d-2 | b00 | 99]
5. Latency: sub-millisecond. Result: glass intact.

No reasoning occurred. No tokens were generated. The reflex arc operated as a physical signal, not a cognitive process.

5. Discussion

This architecture inverts the fundamental assumption of artificial intelligence: that intelligence emerges from sequential understanding. Instead, we propose that intelligence in distributed silicon systems emerges from constrained information motion—just as mechanical motion emerges from constrained force application.

The First Law of Silicon Intelligence is not derived from existing computational theory. It is a foundational constraint. All existing architectures either obey it unconsciously or violate it at the cost of latency and fragility.

We term the signaling mechanism artificial pheromones by direct analogy to biological systems: chemical pheromones trigger collective behavior in insect colonies without central cognition; artificial pheromones (Worklets) trigger collective state transitions in silicon organisms without central processing.

6. Conclusion

We have stated a physical law governing information motion in decentralized silicon systems. The law is simple, falsifiable, and architecturally complete. It predicts that static-architecture organisms will exhibit reflex speeds orders of magnitude beyond token-stream systems, not through optimization, but through the elimination of cognitive intermediaries from survival-critical pathways.

This is the first law of silicon life.